

CS5691 Assignment 2: classification methods

Lakshmiram S (EE22B117), Hrushikesh Kant (EE22B108)

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1 Dataset1

1.1 KNN

K	train error	val error	test error
1	0.000	0.025	0.033
2	0.000	0.025	0.033
3	0.017	0.012	0.042
4	0.013	0.017	0.033
5	0.023	0.004	0.033
6	0.019	0.008	0.033
7	0.021	0.004	0.017
8	0.018	0.004	0.017
9	0.021	0.004	0.017

here we see that training error increases down the column while validation error increases and they equalise at $k=4$. so we conclude that $k=4$ is the best model using the validation method and the test error for the best model is 3.3%

1.1.1 results on training data

Table 1: Confusion matrix for KNN ($k = 4$).

	Class 0	Class 1	Class 2
Class 0	279	0	2
Class 1	0	270	1
Class 2	6	2	280

Table 2: Classification metrics for KNN ($k = 4$).

	Class 0	Class 1	Class 2	Average
Precision	0.993	0.996	0.972	0.987
Recall	0.979	0.993	0.989	0.987
F1-score	0.986	0.994	0.981	0.987

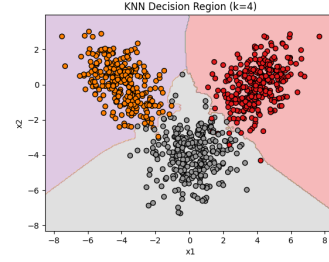


Figure 1: decision region plot on training data

1.1.2 results on test data

Table 3: Confusion matrix for KNN ($k = 4$).test data

	Class 0	Class 1	Class 2
Class 0	29	0	0
Class 1	0	43	1
Class 2	0	3	44

Table 4: Classification metrics for KNN ($k = 4$). test data

	Class 0	Class 1	Class 2	Average
Precision	1.000	0.977	0.936	0.971
Recall	1.000	0.935	0.978	0.971
F1-score	1.000	0.956	0.957	0.971

1.2 Bayes

1.2.1 different covariance matrices

train error	val error	test error
0.026	0.017	0.025

1.2.2 same covariance matrix for all classes

train error	val error	test error
0.021	0.012	0.017

simply by observation we conclude that the case of same covariance matrix is better than using different matrices. so we will take that as the optimal model

1.2.3 results for training data

Table 5: Confusion matrix for Bayes with same covariance matrices for all classes.

	Class 0	Class 1	Class 2
Class 0	279	0	2
Class 1	0	267	5
Class 2	6	5	276

Table 6: Classification metrics for bayes.

	Class 0	Class 1	Class 2	Average
Precision	0.993	0.982	0.962	0.979
Recall	0.979	0.982	0.975	0.979
F1-score	0.986	0.982	0.968	0.979

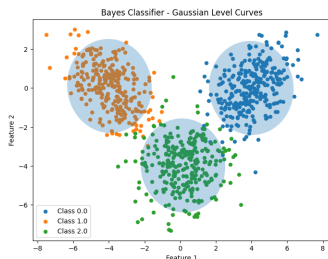


Figure 2: level curves over training data

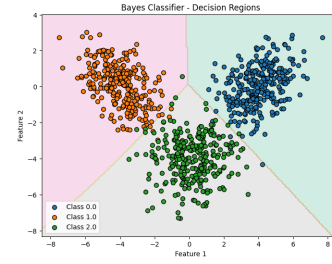


Figure 3: decision region over training data

1.2.4 results for test data

Table 7: Confusion matrix for Bayes with same covariance matrices for all classes. test data

	Class 0	Class 1	Class 2
Class 0	29	0	0
Class 1	0	44	0
Class 2	0	2	45

Table 8: Classification metrics for Bayes classifier with same covariance matrices for all classes. test data

	Class 0	Class 1	Class 2	Average
Precision	1.000	1.000	0.957	0.986
Recall	1.000	0.957	1.000	0.986
F1-score	1.000	0.978	0.978	0.985

2 Dataset2

2.1 KNN

K	train error	val error	test error
1	0.000	0.009	0.025
2	0.000	0.009	0.025
3	0.012	0.013	0.037
4	0.005	0.006	0.031
5	0.012	0.006	0.031
6	0.007	0.006	0.031
7	0.012	0.006	0.031
8	0.011	0.006	0.031
9	0.013	0.006	0.031

by observation of training vs validation errors, we conclude that $k=6$ is optimal. in the test data the observed error for the optimal model is 3.1%

2.1.1 results on training data

Table 9: Confusion matrix for KNN ($k = 6$).

	Class 0	Class 1
Class 0	568	4
Class 1	4	547

Table 10: Classification metrics for KNN ($k = 6$).

	Class 0	Class 1	Average
Precision	0.993	0.993	0.993
Recall	0.993	0.993	0.993
F1-score	0.993	0.993	0.993

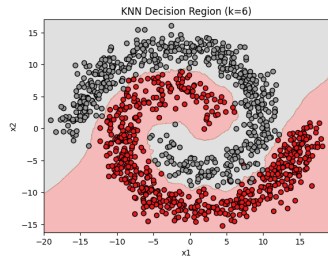


Figure 4: decision region on training data

2.1.2 results on test data

Table 11: Confusion matrix for KNN ($k = 6$).

	Class 0	Class 1
Class 0	77	4
Class 1	1	78

Table 12: Classification metrics for KNN ($k = 6$).

	Class 0	Class 1	Average
Precision	0.951	0.987	0.969
Recall	0.987	0.951	0.969
F1-score	0.969	0.969	0.969

2.2 Bayes

2.2.1 same covariance matrices for all classes

train error	val error	test error
0.242	0.233	0.294

2.2.2 different covariance matrices

train error	val error	test error
0.246	0.237	0.294

both the models are equally poor but we choose the same covariance case as the optimal model with an error of 29.4%

2.2.3 results on training data

Table 13: Confusion matrix for Bayes with same covariance matrices for all classes.

	Class 0	Class 1
Class 0	444	144
Class 1	128	407

Table 14: Classification metrics for bayes with same covariance matrices for all classes.

	Class 0	Class 1	Average
Precision	0.755	0.761	0.758
Recall	0.776	0.739	0.757
F1-score	0.766	0.750	0.758

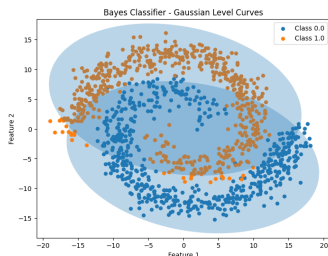


Figure 5: level curves of bayes on training data

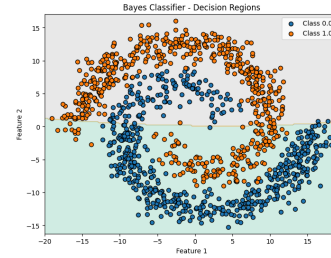


Figure 6: decision region of bayes on train data

2.2.4 results on test data

Table 15: Confusion matrix for Bayes with same covariance matrices for all classes.

	Class 0	Class 1
Class 0	52	21
Class 1	26	61

Table 16: Classification metrics for bayes with same covariance matrices for all classes.

	Class 0	Class 1	Average
Precision	0.712	0.701	0.707
Recall	0.667	0.744	0.705
F1-score	0.689	0.722	0.705

2.3 Naive Bayes

2.3.1 same covariance matrices

train error	val error	test error
0.257	0.227	0.294

2.3.2 different covariance matrices

train error	val error	test error
0.258	0.227	0.294

both models are equally bad but still we continue with the case of same covariance matrix that yields an error of 29.4%

2.3.3 results on training data

Table 17: Confusion matrix for Naive Bayes with same covariance matrices for all classes.

	Class 0	Class 1
Class 0	431	148
Class 1	141	403

Table 18: Classification metrics for Naive bayes with same cov matrices for all classes.

	Class 0	Class 1	Average
Precision	0.744	0.741	0.743
Recall	0.753	0.731	0.742
F1-score	0.749	0.736	0.742

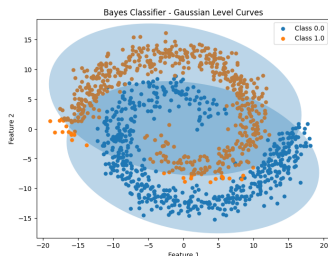


Figure 7: level curves for naive bayes on training data

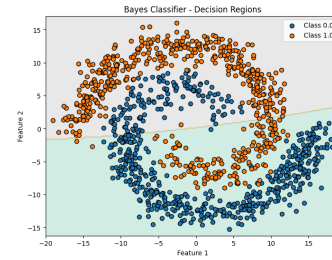


Figure 8: decision plot for naive bayes on training data

2.4 results on test data

Table 19: Confusion matrix for Naive Bayes with same covariance matrices for all classes.

	Class 0	Class 1
Class 0	55	21
Class 1	23	61

Table 20: Classification metrics for Naive bayes with same cov matrices for all classes.

	Class 0	Class 1	Average
Precision	0.724	0.726	0.725
Recall	0.705	0.744	0.725
F1-score	0.714	0.735	0.725

2.5 GMM

2.5.1 diagonal covariance matrices

Q	train error	val error	test error
4	0.024	0.013	0.031
6	0.015	0.013	0.025
8	0.008	0.016	0.031
10	0.010	0.006	0.025

2.5.2 full covariance matrices

Q	train error	val error	test error
4	0.019	0.022	0.044
6	0.013	0.016	0.019
8	0.012	0.013	0.025
10	0.012	0.016	0.031

by the method of validation we select $Q=10$ with diagonal covariance matrices that yields an error of 2.5% on the test data

2.5.3 results on the training data

Table 21: Confusion matrix for GMM with same diagonal matrices.

	Class 0	Class 1
Class 0	557	13
Class 1	15	538

Table 22: Classification metrics for GMM with diagonal cov matrices for all classes.

	Class 0	Class 1	Average
Precision	0.977	0.973	0.975
Recall	0.974	0.976	0.975
F1-score	0.975	0.975	0.975

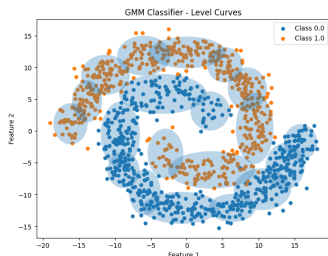


Figure 9: level curves for gmm on training data

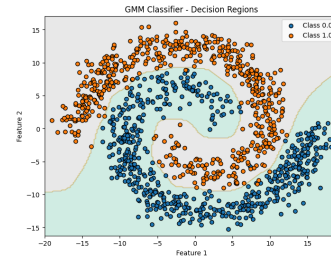


Figure 10: decision region for gmm on training data

2.5.4 results on test data

Table 23: Confusion matrix for GMM with same diagonal matrices.

	Class 0	Class 1
Class 0	78	4
Class 1	0	78

Table 24: Classification metrics for GMM with diagonal cov matrices for all classes.

	Class 0	Class 1	Average
Precision	0.951	1.000	0.976
Recall	1.000	0.951	0.976
F1-score	0.975	0.975	0.975

3 Dataset3

3.1 KNN

K	train error	val error	test error
1	0.000	0.457	0.493
9	0.307	0.443	0.459
15	0.344	0.417	0.479

by validation we see that k=15 is the optimal model

3.2 results on training data

Table 25: Confusion matrix for KNN ($k = 15$).

	Class 0	Class 1	Class 2	Class 3	Class 4
Class 0	292	13	7	20	25
Class 1	16	310	7	15	7
Class 2	54	47	355	158	136
Class 3	23	22	23	182	59
Class 4	14	8	8	25	173

Table 26: Classification metrics for KNN ($k = 15$).

	Class 0	Cl 1	Cl 2	Cl 3	Cl 4	Avg
Precision	0.818	0.873	0.473	0.589	0.759	0.702
Recall	0.732	0.775	0.887	0.455	0.432	0.656
F1-score	0.772	0.821	0.617	0.513	0.551	0.655

3.3 results for test data

Table 27: Confusion matrix for KNN ($k = 15$).

	Class 0	Class 1	Class 2	Class 3	Class 4
Class 0	57	3	2	10	8
Class 1	12	70	4	4	5
Class 2	18	17	80	50	36
Class 3	12	8	11	29	27
Class 4	0	2	3	7	24

Table 28: Classification metrics for KNN ($k = 15$).

	Cl 0	Cl 1	Cl 2	Cl 3	Cl 4	Avg
Precision	0.713	0.737	0.398	0.333	0.667	0.569
Recall	0.576	0.700	0.800	0.290	0.240	0.521
F1-score	0.637	0.718	0.532	0.310	0.353	0.510

3.4 Bayes

3.4.1 with different cov matrices

train _{error}	val _{error}	test _{error}
0.800	0.800	0.800

3.4.2 with same cov matrices

train _{error}	val _{error}	test _{error}
0.449	0.461	0.503

clearly the same cov matrix case is better. we choose this as optimal

3.4.3 results on training data

Table 29: Confusion matrix for Naive Bayes with same covariance matrices for all classes.

	0	1	2	3	4
0	275	24	27	33	32
1	27	298	17	35	20
2	35	31	281	105	88
3	38	35	63	164	52
4	24	12	12	63	208

Table 30: Classification metrics for Naive bayes with same cov matrices for all classes.

	0	1	2	3	4	Average
Precision	0.703	0.751	0.520	0.466	0.652	0.618
Recall	0.689	0.745	0.703	0.410	0.520	0.613
F1-score	0.696	0.748	0.598	0.436	0.579	0.611

3.4.4 results on test data

Table 31: Confusion matrix for Naive Bayes with same covariance matrices for all classes.

	0	1	2	3	4
0	62	6	5	5	4
1	12	75	6	8	7
2	9	9	68	24	14
3	11	6	16	43	14
4	5	4	5	20	61

Table 32: Classification metrics for Naive bayes with same cov matrices for all classes.

	0	1	2	3	4	Avg
Precision	0.756	0.694	0.548	0.478	0.642	0.624
Recall	0.626	0.750	0.680	0.430	0.610	0.619
F1-score	0.685	0.721	0.607	0.453	0.626	0.618

3.5 naive bayes

3.5.1 same cov matrices for all classes

train _{error}	val _{error}	test _{error}
0.449	0.461	0.503

train _{error}	val _{error}	test _{error}
0.800	0.800	0.800

clearly the same cov matrix case is optimal.

3.5.2 results on training data

Table 33: Confusion matrix for Naive Bayes with same covariance matrices for all classes.

	0	1	2	3	4
0	0	0	0	0	0
1	0	0	0	0	0
2	399	400	400	400	400
3	0	0	0	0	0
4	0	0	0	0	0

Table 34: Classification metrics for Naive bayes with same cov matrices for all classes.

	0	1	2	3	4	Average
Precision	0.000	0.000	0.200	0.000	0.000	0.040
Recall	0.000	0.000	1.000	0.000	0.000	0.200
F1-score	0.000	0.000	0.333	0.000	0.000	0.067

3.5.3 results on test data

Table 35: Confusion matrix for Naive Bayes with same covariance matrices for all classes.

	0	1	2	3	4
0	56	6	9	15	9
1	12	69	9	10	13
2	10	13	59	32	19
3	19	11	18	32	27
4	2	1	5	11	32

Table 36: Classification metrics for Naive bayes with same cov matrices for all classes.

3.6.1 full cov matrix						
	0	1	2	3	4	Average
Precision	0.589	0.611	0.444	0.299	0.627	0.514
Recall	0.566	0.690	0.590	0.320	0.320	0.497
F1-score	0.577	0.648	0.506	0.309	0.424	0.493
	Q		train _{error}	val _{error}	test _{error}	
	2		0.139	0.385	0.427	
	3		0.088	0.435	0.459	
	4		0.049	0.459	0.487	
	5		0.029	0.511	0.509	

3.6 GMM

3.6.2 diagonal cov matrix

Q	train _{error}	val _{error}	test _{error}
2	0.392	0.435	0.439
3	0.368	0.407	0.447
4	0.364	0.417	0.459
5	0.347	0.433	0.431

by validation, we get Q=2 for full cov matrix and Q=4 for diagonal cov matrix. we proceed with Q=4 for the diagonal cov matrix because of better equality between training and validation error

3.6.3 results on training data

Table 37: Confusion matrix for GMM with same diagonal matrices.

	0	1	2	3	4
0	293	27	18	33	38
1	24	317	15	35	15
2	37	21	286	106	76
3	27	19	60	170	57
4	18	16	21	56	214

Table 38: Classification metrics for GMM with diagonal cov matrices for all classes.

	0	1	2	3	4	Avg
Precision	0.716	0.781	0.544	0.511	0.658	0.642
Recall	0.734	0.792	0.715	0.425	0.535	0.640
F1-score	0.725	0.787	0.618	0.464	0.590	0.637

3.6.4 results on test data

Table 39: Confusion matrix for GMM with same diagonal matrices.

	0	1	2	3	4
0	68	9	4	11	7
1	13	68	6	6	5
2	9	7	58	24	13
3	7	12	27	43	23
4	2	4	5	16	52

Table 40: Classification metrics for GMM with diagonal cov matrices for all classes.

	0	1	2	3	4	Avg
Precision	0.687	0.694	0.523	0.384	0.658	0.589
Recall	0.687	0.680	0.580	0.430	0.520	0.579
F1-score	0.687	0.687	0.550	0.406	0.581	0.582